ABW501 Analytics Edge

Trendlines and Regression Analysis



Descriptive Analytics

Predictive Analytics

Prescriptive Analytics



Learning Objective

- Explain purpose of regression analysis and provide examples in business
- Use scatter chart to identify the type of relationship between two variables
- Explain how least-squares regression finds the best-fitting regression model
- Interpret both linear and multiple regression statistics output
- Draw conclusions for tests of hypotheses about regression coefficients
- Apply systematic approach to build good regression models

Learning Objective

- Explain the importance of understanding multicollinearity in regression models
- Build regression models for categorical data using dummy variables
- Test for interactions in regression models with categorical variables
- Identify when curvilinear regression models are more appropriate than linear models



Modelling Relationships and Trends in Data

Modelling Relationships and Trends in Data

Create charts to better understand data sets

- For cross-sectional data, use a scatter chart
- For time series data, use a line chart

Common Mathematical Functions Used

- Linear $y = a + bx$
- Logarithmic $y = \ln(x)$
- Polynomial (2nd order) $y = ax^2 + bx + c$
- Polynomial (3rd order) $y = ax^3 + bx^2 + dx + e$
- Power $y = ax^b$
- Exponential $y = ab^x$

(the base of natural logarithms, e = 2.71828…is often used for the constant b)

R^2

R^2 (R-squared) is a measure of the "fit" of the line to the data

- The value of R^2 will be between 0 and 1.
- A value of 1.0 indicates a perfect fit and all data points would lie on the line
- The larger the value of R^2 the better the fit.

Modelling a Price-Demand Function

Example: Market research study collected data on sales volumes for different levels of pricing for a particular product.

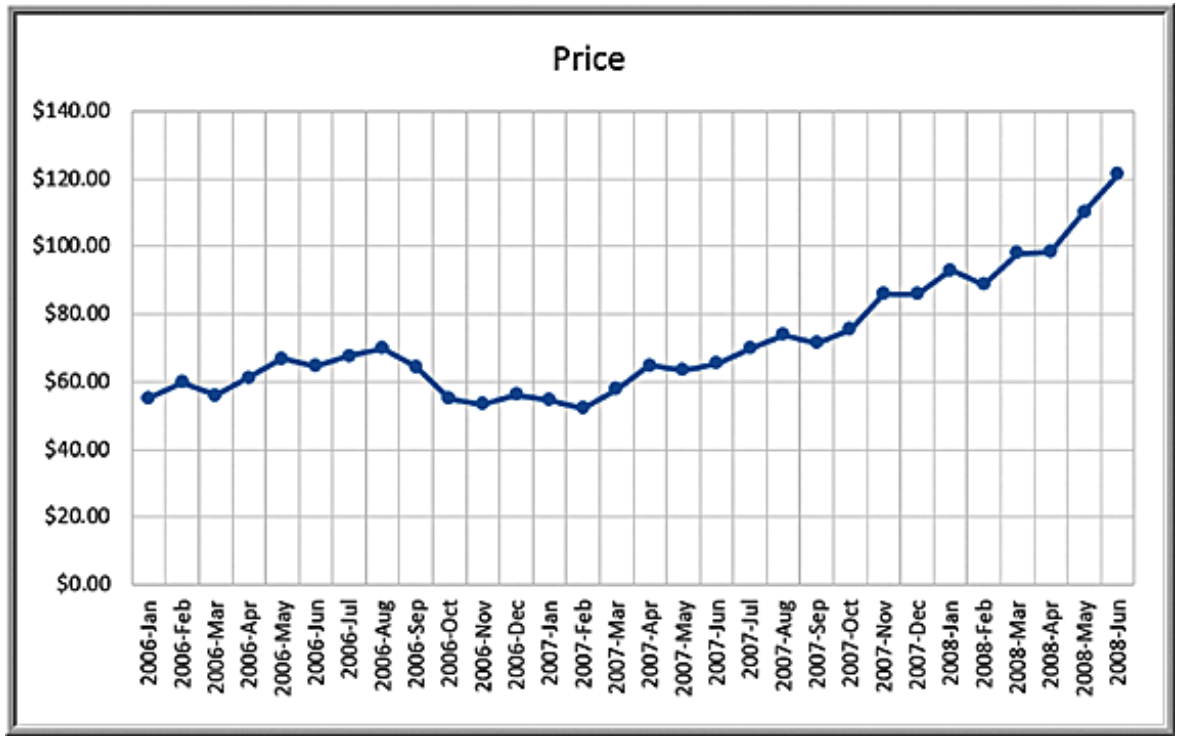
Linear demand function

$$\text{Sales} = 20{,}512 - 9.5117(\text{price})$$



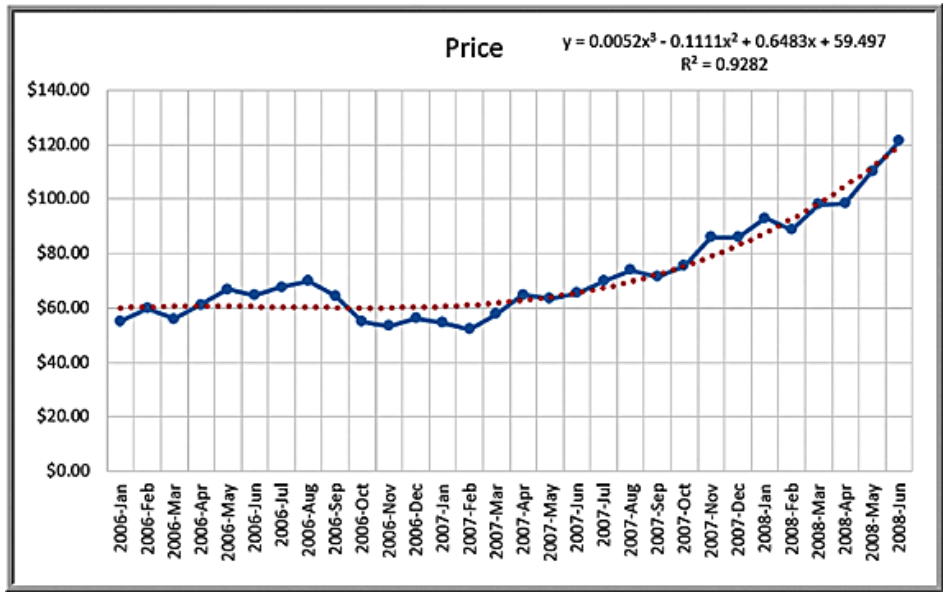
Predicting Crude Oil Prices

Example: A chart of historical data on crude oil prices each month from January 2006 to Jun 2008. Using trendline, we can try to fit the various functions to these data.



Predicting Crude Oil Prices

Exponential	$y = 50.49e^{0.021x}$	$R^2 = 0.664$
Logarithmic	$y = 13.02\ln(x) + 39.60$	$R^2 = 0.382$
Polynomial 2°	$y = 0.13x^2 - 2.399x + 68.01$	$R^2 = 0.905$
Polynomial 3°	$y = 0.005x^3 - 0.111x^2 + 0.648x + 59.497$	$R^2 = 0.928$ *
Power	$y = 45.96x^{0.0169}$	$R^2 = 0.397$



*Best fitting line – 3rd order polynomials

Caution about Polynomials

- The R^2 value will continue to increase as the order of the polynomial increases; that is, a 4th order polynomial will provide a better fit than a 3rd order, and so on.
- Higher order polynomials will generally not be very smooth and will be difficult to interpret visually.
- Thus, it is not recommend to go beyond a third-order polynomial when fitting data.
- Use your eye to make a good judgment!



Simple Linear Regression

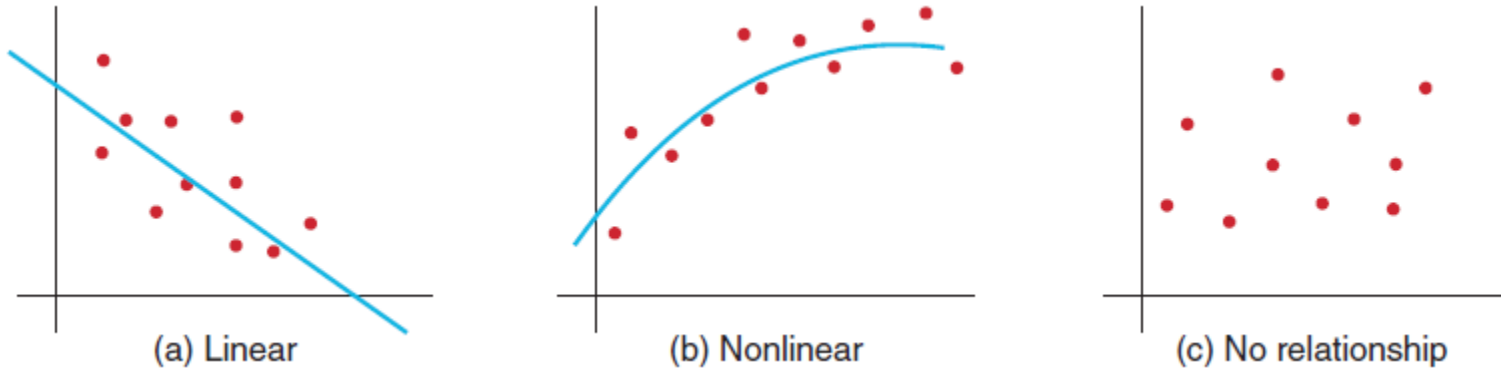
Regression Analysis

Regression analysis is a tool for building mathematical and statistical models that characterize relationships between a dependent (ratio) variable and one or more independent, or explanatory variables (ratio or categorical), all of which are numerical.

- **Simple linear regression** involves a single independent variable.
- **Multiple regression** involves two or more independent variables.

Simple Linear Regression

- Finds a linear relationship between:
  - one independent variable X and
  - one dependent variable Y
- First prepare a scatter plot to verify the data has a linear trend.
- Use alternative approaches if the data is not linear.



Home Market Value Data

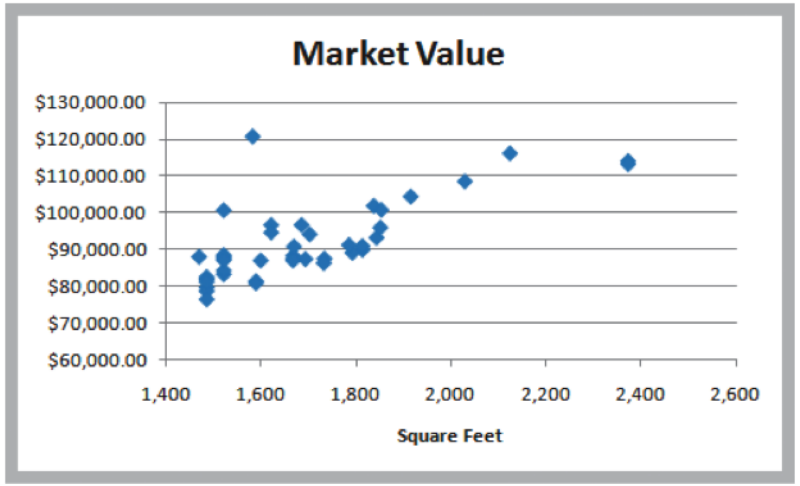
Example: Size of a house is typically related to its market value.

X =square footage

Y =market value ($)

The scatter plot of the full data set (42 homes) indicates a linear trend.

	A	B	C
1	Home Market Value		
2			
3	House Age	Square Feet	Market Value
4	33	1,812	$90,000.00
5	32	1,914	$104,400.00
6	32	1,842	$93,300.00
7	33	1,812	$91,000.00
8	32	1,836	$101,900.00
9	33	2,028	$108,500.00
10	32	1,732	$87,600.00

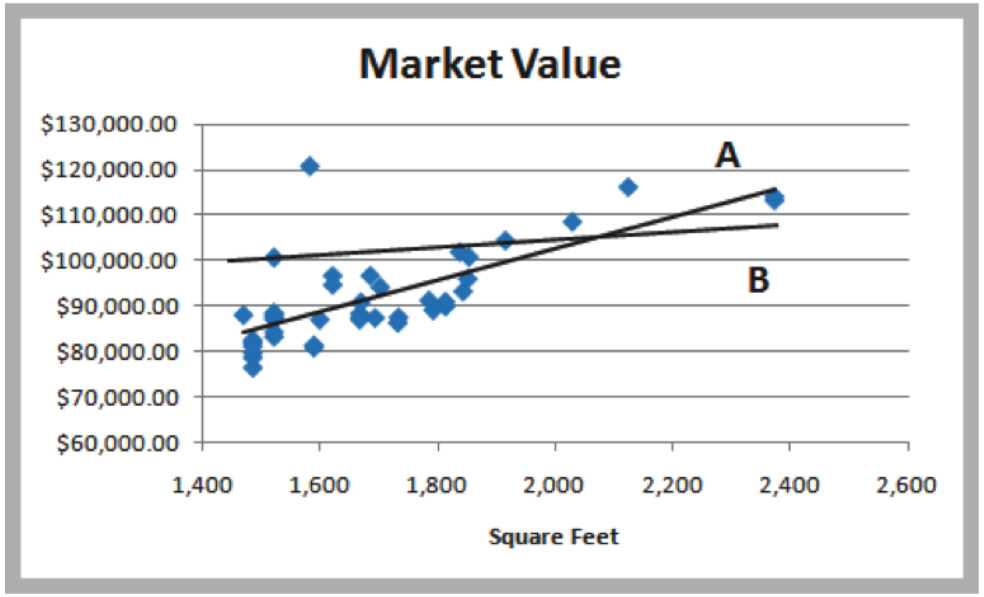


Finding the Best-Fitting Regression Line

- Express the relationship between the dependent and independent variables by a single linear equation

$$\text{Market value} = a + b * \text{square feet}$$

- Two possible straight lines, A and B
- Line A is clearly a better fit to the data
- We want to determine the best regression line

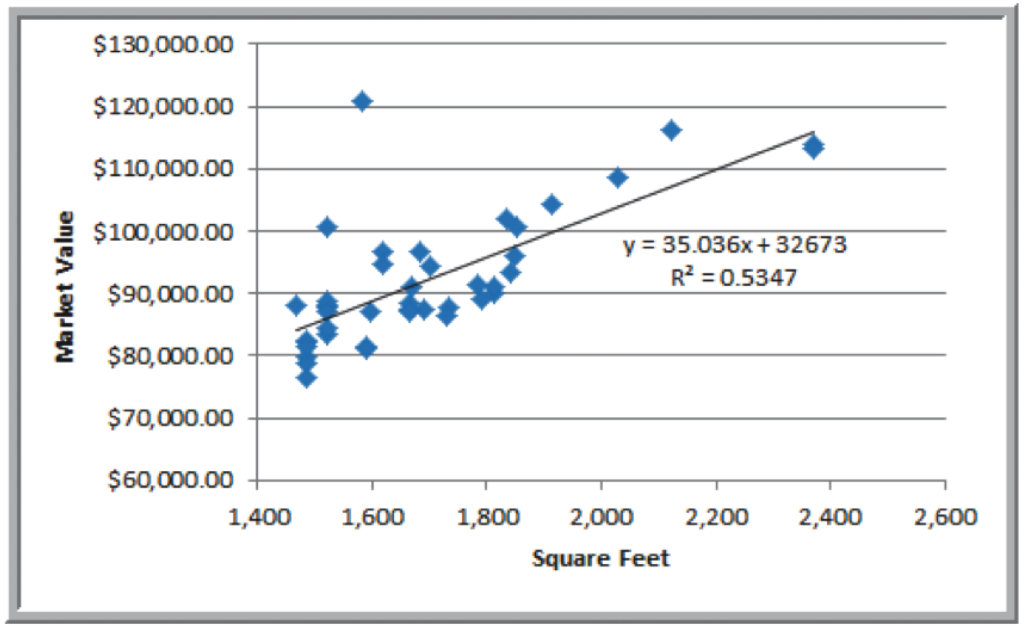


Finding the Best-Fitting Regression Line

$$\text{Market value} = 32{,}673 + 35.036 * \text{square feet}$$

The estimated market value of a 2,200 square feet home would be:

$$\text{Market value} = 32{,}673 + 35.036 * 2{,}200 = \$109{,}752$$



The regression model explains variation in market value due to size of the home. If provides better estimates of market value than simply using the average.

Least-Squares Regression

- Simple linear regression model:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

- We estimate the parameters from the sample data:

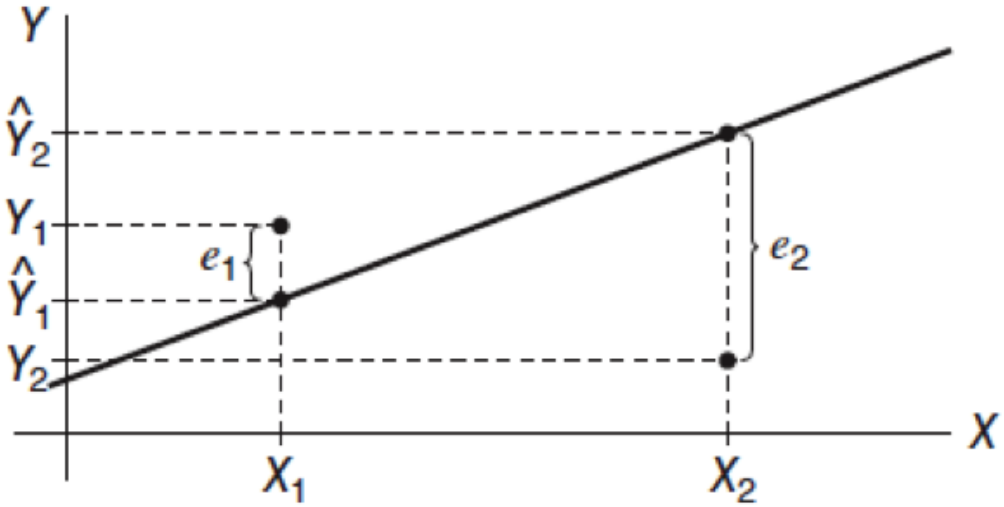
$$\hat{Y} = b_0 + b_1 X$$

- Let X_i be the value of the independent variable of the ith observation. When the value of the independent variable is X_i, then $Y_i = b_0 + b_1 X_i$ is the estimated value of Y for X_i.

Residuals

- Residuals are the observed errors associated with estimating the value of the dependent variable using the regression line:

$$e_i = Y_i - \hat{Y}_i$$



Errors associated with individual observations

Least Squares Regression

- The best-fitting line minimizes the sum of squares of the residuals.

$$\sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2 = \sum_{i=1}^{n} (Y_i - [b_0 + b_1 X_i])^2$$

$$b_1 = \frac{\sum_{i=1}^{n} X_i Y_i - n\overline{X}\,\overline{Y}}{\sum_{i=1}^{n} X_i^2 - n\overline{X}^2}$$

$$b_0 = \overline{Y} - b_1 \overline{X}$$

Regression Statistics

- **Multiple R** – $|r|$, where r is the sample correlation coefficient. The value of r varies from -1 to $+1$ (r is negative if slope is negative)
- **R Square** – coefficient of determination, R^2, which varies from 0 (no fit) to 1 (perfect fit)
- **Adjusted R Square** – adjusts R^2 for sample size and number of X variables
- **Standard Error** – variability between observed and predicted Y values. This is formally called the **standard error of the estimate,** $\boldsymbol{S_{YX}}$.

Interpreting Regression Statistics for Simple Linear Regression

	A	B	C	D	E	F	G
1	Regression Analysis						
2							
3	*Regression Statistics*						
4	Multiple R	0.731255223					
5	R Square	0.534734202					
6	Adjusted R Square	0.523102557					
7	Standard Error	7287.722712					
8	Observations	42					
9							
10	ANOVA						
11		*df*	*SS*	*MS*	*F*	*Significance F*	
12	Regression	1	2441633669	2441633669	45.97236277	3.79802E-08	
13	Residual	40	2124436093	53110902.32			
14	Total	41	4566069762				
15							
16		*Coefficients*	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
17	Intercept	32673.2199	8831.950745	3.699434116	0.000649604	14823.18178	50523.25802
18	Square Feet	35.03637258	5.16738385	6.780292234	3.79802E-08	24.59270036	45.48004481

53% of the variation in home market values can be explained by home size.

The standard error of $7287 is less than standard deviation (not shown) of $10,553.

Regression as Analysis of Variance

- ANOVA conducts an F-test to determine whether variation in Y is due to varying levels of X.
- ANOVA is used to test for significance of regression:
- H_0: population slope coefficient $= 0$
- H_1: population slope coefficient $\neq 0$
- Rejecting H_0 indicates that X explains variation in Y.

Interpreting Significance of Regression

Example: From *Home Market Value*

H_0: $\beta_1 = 0$ Home size is NOT a significant variable

H_0: $\beta_1 \neq 0$ Home size is a significant variable

p-value= $3.798\text{x}10^{-8}$

Reject H_0. The slope is not equal to zero. Using linear relationship, home size is a significant variable in explaining variation in market value.

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Testing Hypotheses for Regression Coefficient

An alternate method for testing whether a slope or intercept is zero is to use a t-test:

$$t = \frac{b_1 - 0}{\text{standard error}}$$

Use p-values to draw conclusion.

Neither coefficient is statistically equal to zero

	A	B	C	D	E	F	G
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Confidence Intervals for Regression Coefficients

- Confidence intervals (Lower 95% and Upper 95% values in the output) provide information about the unknown values of the true regression coefficients, accounting for sampling error.
- We may also use confidence intervals to test hypotheses about the regression coefficients.
- To test the hypotheses

$$H_0: \beta_1 = B_1$$
$$H_1: \beta_1 \neq B_1$$

check whether B_1 falls within the confidence interval for the slope. If it does, reject the null hypothesis

Interpreting Confidence Interval for Regression Coefficients

Example: For the Home Market Value data, a 95% confidence interval for the intercept is [14,823, 50,523], and for the slope, [24.59, 45.48].

Although we estimated that a house with 1,750 square feet has a market value of 32,673 + 35.036(1,750) =$93,986, if the true population parameters are at the extremes of the confidence intervals, the estimate might be as low as 14,823 + 24.59(1,750) = $57,855 or as high as 50,523 + 45.48(1,750) = $130,113.

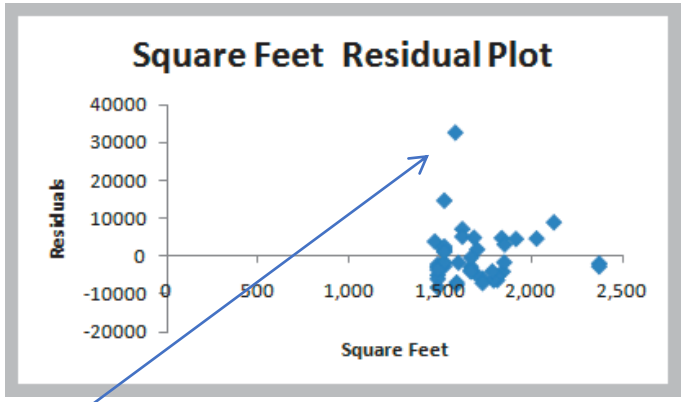


Residual Analysis and Regression Assumptions

Residual Analysis

- **Residual** = Actual Y value – Predicted Y value
- **Standard residual** = residual / standard deviation
- Rule of thumb: Standard residuals outside of ±2 or ±3 are potential outliers.

	A	B	C	D
22	RESIDUAL OUTPUT			
23				
24	*Observation*	*Predicted Market Value*	*Residuals*	*Standard Residuals*
25	1	96159.12702	-6159.127018	-0.855636403
26	2	99732.83702	4667.162978	0.64837022
27	3	97210.2182	-3910.218196	-0.543214164
28	4	96159.12702	-5159.127018	-0.716714702
29	5	96999.99996	4900.00004	0.680716341



This point has a standard residual of 4.53

Checking Assumptions

Linearity

- examine scatter diagram (should appear linear)
- examine residual plot (should appear random)

Normality of Errors

- view a histogram of standard residuals
- regression is robust to departures from normality

Homoscedasticity: variation about the regression line is constant

- examine the residual plot

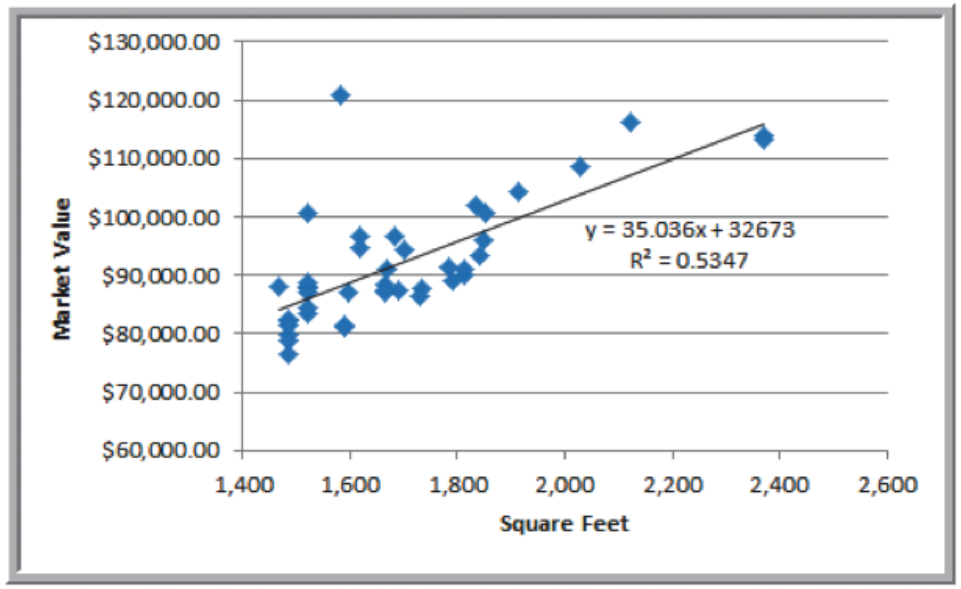
Independence of Errors: successive observations should not be related.

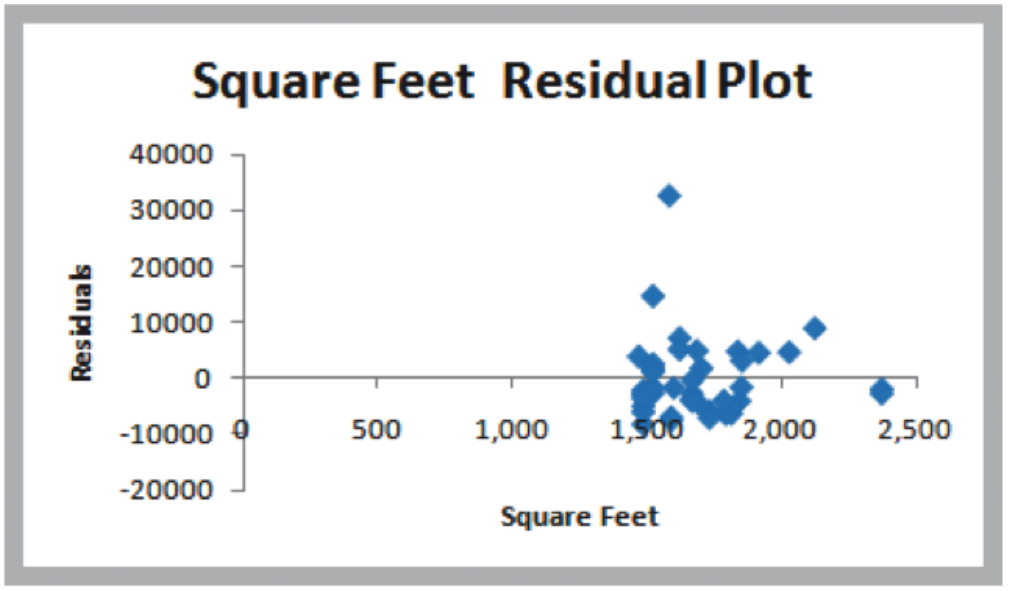
- This is important when the independent variable is time.

Example: Checking Regression Assumptions

Linearity - linear trend in scatterplot

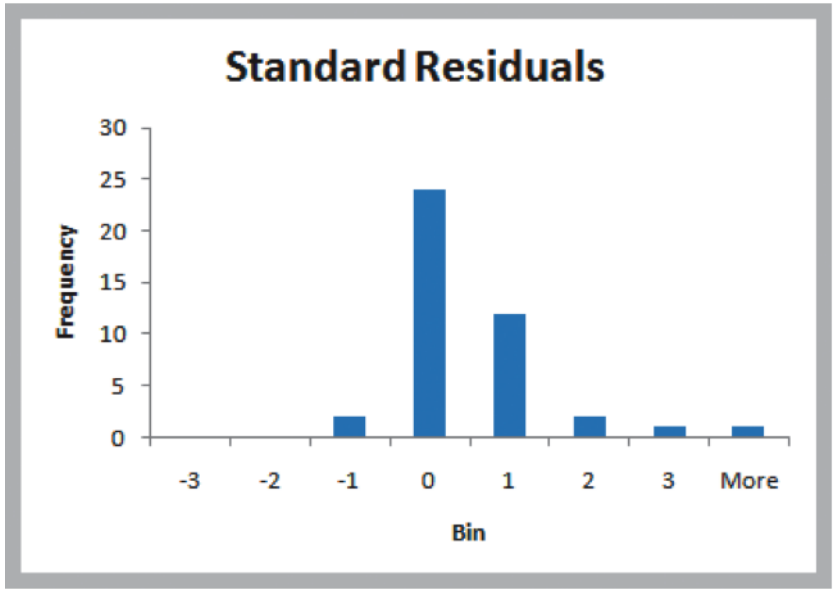
- no pattern in residual plot





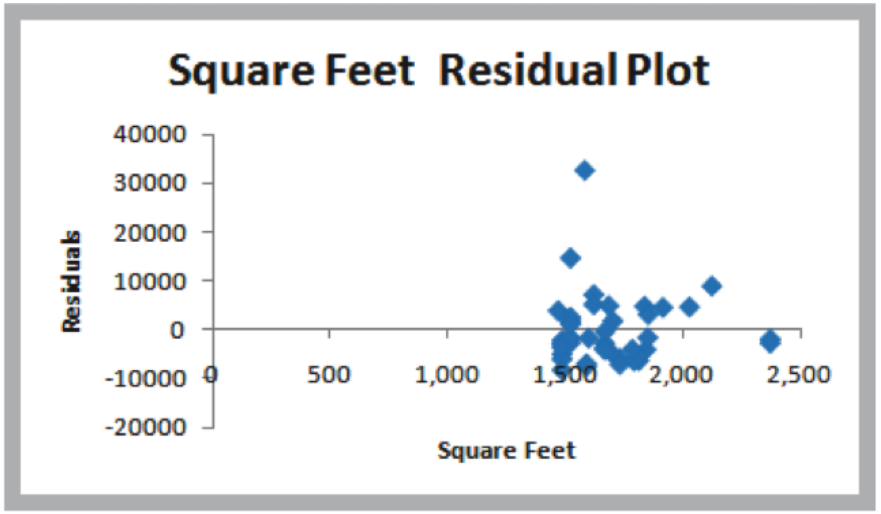
Example: Checking Regression Assumptions

Normality of Errors – residual histogram appears slightly skewed but is not a serious departure



Example: Checking Regression Assumptions

Homoscedasticity – residual plot shows no serious difference in the spread of the data for different X values



Independence of Errors – Because the data is cross-sectional, we can assume this assumption holds.



Multiple Linear Regression

Multiple Linear Regression

- A linear regression model with more than one independent variable is called a **multiple linear regression model**

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \varepsilon$$

where

Y is the dependent variable,
$X_1, \ldots, X_k$ are the independent (explanatory) variables,
β_0 is the intercept term,
$\beta_1, \ldots, \beta_k$ are the regression coefficients for the independent variables,
ε is the error term

Estimated Multiple Regression Equation

- We estimate the regression coefficients—called **partial regression coefficients** — $b_0, b_1, b_2, \ldots b_k$, then use the model:

$$\hat{Y} = b_0 + b_1X_1 + b_2X_2 + \cdots + b_kX_k$$

- The partial regression coefficients represent the expected change in the dependent variable when the associated independent variable is increased by one unit while the values of all other independent variables are held constant.

ANOVA for Multiple Regression

- ANOVA tests for significance of the entire model. That is, it computes an F-statistic for testing the hypotheses:

$$H_0: \beta_1 = \beta_2 = \cdots = \beta_k = 0$$

$$H_1: \text{at least one } \beta_j \text{ is not } 0$$

- The multiple linear regression output also provides information to test hypotheses about each of the individual regression coefficients.
  - If we reject the null hypothesis that the slope associated with independent variable i is 0, then the independent variable i is significant and improves the ability of the model to better predict the dependent variable. If we cannot reject H_0, then that independent variable is not significant and probably should not be included in the model.

Interpreting Regression Results

Example: The ***Colleges and Universities Data*** are used to predict student graduation rates using several indicators.

	A	B	C	D	E	F	G
1	**Colleges and Universities**						
2							
3	**School**	**Type**	**Median SAT**	**Acceptance Rate**	**Expenditures/Student**	**Top 10% HS**	**Graduation %**
4	Amherst	Lib Arts	1315	22%	$ 26,636	85	93
5	Barnard	Lib Arts	1220	53%	$ 17,653	69	80
6	Bates	Lib Arts	1240	36%	$ 17,554	58	88
7	Berkeley	University	1176	37%	$ 23,665	95	68
8	Bowdoin	Lib Arts	1300	24%	$ 25,703	78	90
9	Brown	University	1281	24%	$ 24,201	80	90

Interpreting Regression Results

	A	B	C	D	E	F	G
1	SUMMARY OUTPUT						
2							
3	*Regression Statistics*						
4	Multiple R	0.731044486					
5	R Square	0.534426041					
6	Adjusted R Square	0.492101135					
7	Standard Error	5.30833812					
8	Observations	49					
9							
10	ANOVA						
11		*df*	*SS*	*MS*	*F*	*Significance F*	
12	Regression	4	1423.209266	355.8023166	12.62675098	6.33158E-07	
13	Residual	44	1239.851958	28.1784536			
14	Total	48	2663.061224				
15							
16		*Coefficients*	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
17	Intercept	17.92095587	24.55722367	0.729763108	0.469402466	-31.57087643	67.41278818
18	Median SAT	0.072006285	0.017983915	4.003927007	0.000236106	0.035762085	0.108250485
19	Acceptance Rate	-24.8592318	8.315184822	-2.989618672	0.004559569	-41.61738567	-8.101077939
20	Expenditures/Student	-0.00013565	6.59314E-05	-2.057438385	0.045600178	-0.000268526	-2.77379E-06
21	Top 10% HS	-0.162764489	0.079344518	-2.051364015	0.046213848	-0.322672857	-0.00285612

$$\text{Graduation\%} = 17.92 + 0.072\,\text{SAT} - 24.859\,\text{ACCEPTANCE} - 0.000136\,\text{EXPENDITURES} - 0.163\,\text{TOP10\% HS}$$

- The value of R^2 indicates that 53% of the variation in the dependent variable is explained by these independent variables.
- All coefficients are statistically significant.



Building Good Regression Models

Model Building Issues

- A good regression model should include only significant independent variables.
- However, it is not always clear exactly what will happen when we add or remove variables from a model; variables that are (or are not) significant in one model may (or may not) be significant in another.
  - Therefore, you should not consider dropping all insignificant variables at one time, but rather take a more structured approach.
- Adding an independent variable to a regression model will always result in R^2 equal to or greater than the R^2 of the original model.
- Adjusted R^2 reflects both the number of independent variables and the sample size and may either increase or decrease when an independent variable is added or dropped. An increase in adjusted R^2 indicates that the model has improved.

Systematic Model Building Approach

① Construct a model with all available independent variables. Check for significance of the independent variables by examining the p-values

② Identify the independent variable having the largest p-value that exceeds the chosen level of significance

③ Remove the variable identified in ② from the model and evaluate adjusted R^2

(Don't remove all variables with p-values that exceed α at the same time, but remove only one at a time)

④ Continue until all variables are significant

Identifying the Best Regression Model

Example: ① Identify variable with the largest p-value exceeding 0.05. ② In this case, it is *Home Value*. ③ Drop from the model and rerun the regression.

	A	B	C	D	E	F	G
1	SUMMARY OUTPUT						
2							
3	*Regression Statistics*						
4	Multiple R	0.97309221					
5	R Square	0.946908448					
6	Adjusted R Square	0.944143263					
7	Standard Error	2055.64333					
8	Observations	102					
9							
10	ANOVA						
11		*df*	*SS*	*MS*	*F*	*Significance F*	
12	Regression	5	7235179873	1447035975	342.4394584	1.5184E-59	
13	Residual	96	405664271.9	4225669.499			
14	Total	101	7640844145				
15							
16		*Coefficients*	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
17	Intercept	-10710.64278	4260.976308	-2.513659314	0.013613179	-19168.61391	-2252.671659
18	Age	318.6649626	60.98611242	5.225205378	1.01152E-06	197.6084862	439.721439
19	Education	621.8603472	318.9595184	1.949652891	0.054135377	-11.26929279	1254.989987
20	Income	0.146323453	0.040781001	3.588029937	0.000526666	0.065373806	0.227273101
21	Home Value	0.009183067	0.011038075	0.831944635	0.407504891	-0.012727338	0.031093473
22	Wealth	0.074331533	0.011189265	6.643111131	1.84838E-09	0.052121017	0.096542049

Identifying the Best Regression Model

Example: Regression after removing *Home Value*. All p-values are now less than 0.05, so this appears to be the best model.

	A	B	C	D	E	F	G
1	SUMMARY OUTPUT						
2							
3	*Regression Statistics*						
4	Multiple R	0.97289551					
5	R Square	0.946525674					
6	Adjusted R Square	0.944320547					
7	Standard Error	2052.378536					
8	Observations	102					
9							
10	ANOVA						
11		*df*	*SS*	*MS*	*F*	*Significance F*	
12	Regression	4	7232255152	1808063788	429.2386497	9.68905E-61	
13	Residual	97	408588992.5	4212257.655			
14	Total	101	7640844145				
15							
16		*Coefficients*	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
17	Intercept	-12432.45673	3718.674319	-3.34324968	0.001177705	-19812.99587	-5051.917589
18	Age	325.0652837	60.40284468	5.381622098	5.1267E-07	205.1823574	444.9482101
19	Education	773.3800418	261.4330936	2.958233142	0.003886994	254.5077194	1292.252364
20	Income	0.159747379	0.037393587	4.272052794	4.52422E-05	0.085531459	0.233963298
21	Wealth	0.072988791	0.011054665	6.602532898	2.16051E-09	0.051048341	0.094929242

Adjusted R^2 improves slightly.
All *X* variables are significant.

Alternate Criterion

- Use the t-statistic.
- If $|t| < 1$, then standard error will decrease and adjusted R^2 will increase if the variable is removed. If $| t | > 1$, then the opposite will occur.
- You can follow the same systematic approach, except using t-values instead of p-values

Multicollinearity

- Occurs when there are strong correlations among the independent variables, and they can predict each other better than the dependent variable.
- When significant multicollinearity is present, it becomes difficult to isolate the effect of one independent variable on the dependent variable, the signs of coefficients may be the opposite of what they should be, making it difficult to interpret regression coefficients, and p-values can be inflated.
- Correlations exceeding ±0.7 may indicate multicollinearity
- The **variance inflation factor** is a better indicator

Identifying Potential Multicollinearity

Colleges and Universities correlation matrix; none exceed the recommended threshold of ± 0.7

	A	B	C	D	E	F
1		*Median SAT*	*Acceptance Rate*	*Expenditures/Student*	*Top 10% HS*	*Graduation %*
2	Median SAT	1				
3	Acceptance Rate	-0.601901959	1			
4	Expenditures/Student	0.572741729	-0.284254415	1		
5	Top 10% HS	0.503467995	-0.609720972	0.505782049	1	
6	Graduation %	0.564146827	-0.55037751	0.042503514	0.138612667	1

Banking Data correlation matrix; large correlations exist

	A	B	C	D	E	F	G
1		*Age*	*Education*	*Income*	*Home Value*	*Wealth*	*Balance*
2	Age	1					
3	Education	0.173407147	1				
4	Income	0.4771474	0.57539402	1			
5	Home Value	0.386493114	0.753521067	0.795355158	1		
6	Wealth	0.468091791	0.469413035	0.946665447	0.698477789	1	
7	Balance	0.565466834	0.55488066	0.951684494	0.766387128	0.948711734	1

Identifying Potential Multicollinearity

1. If we remove *Wealth* from the model, the adjusted R^2 drops to 0.9201, but we discover that *Education* is no longer significant
2. Dropping *Education* and leaving only *Age* and *Income* in the model results in adjusted R^2 of 0.9202
3. However, if we remove *Income* from the model instead of *Wealth*, the adjusted R^2 drops only to 0.9345, and all remaining variables (*Age*, *Education*, and *Wealth*) are significant.

	A	B	C	D	E	F	G
1	SUMMARY OUTPUT						
2							
3	*Regression Statistics*						
4	Multiple R	0.967710981					
5	R Square	0.936464543					
6	Adjusted R Square	0.93451958					
7	Standard Error	2225.695322					
8	Observations	102					
9							
10	ANOVA						
11		*df*	*SS*	*MS*	*F*	*Significance F*	
12	Regression	3	7155379617	2385126539	481.4819367	1.71667E-58	
13	Residual	98	485464527.3	4953719.667			
14	Total	101	7640844145				
15							
16		*Coefficients*	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
17	Intercept	-17732.45142	3801.662822	-4.664393517	9.79978E-06	-25276.72757	-10188.17528
18	Age	367.8214086	64.59823831	5.693985134	1.2977E-07	239.6283071	496.0145102
19	Education	1300.308712	249.9731413	5.201793703	1.08292E-06	804.2451489	1796.372276
20	Wealth	0.116467903	0.004679827	24.88722652	3.75813E-44	0.107180939	0.125754866

Practical Issues in Trendline and Regression Modelling

- Identifying the best regression model often requires experimentation and trial and error.
- The independent variables selected should make sense in attempting to explain the dependent variable
- Logic should guide your model development. In many applications, behavioural, economic, or physical theory might suggest that certain variables should belong in a model.

Practical Issues in Trendline and Regression Modelling

- Additional variables increase R^2 and therefore, help to explain a larger proportion of the variation.
- Even though a variable with a large p-value is not statistically significant, it could simply be the result of sampling error and a modeler might wish to keep it.
- Good models are as simple as possible (the principle of **parsimony**).

Overfitting

- Overfitting means fitting a model too closely to the sample data at the risk of not fitting it well to the population in which we are interested.
- While fitting data with higher-order polynomial functions might provide a better mathematical fit to the sample data, doing so can make it difficult to explain the phenomena rationally.
- In multiple regression, if we add too many terms to the model, then the model may not adequately predict other values from the population.
- Overfitting can be mitigated by using good logic, intuition, theory, and parsimony



Regression with Categorical Variables

Regression with Categorical Variables

- Regression analysis requires numerical data.
- Categorical data can be included as independent variables, but must be coded numeric using dummy variables.
- For variables with 2 categories, code as 0 and 1

A Model with Categorical Variables

Example: ***Employee Salaries*** provides data for 35 employees

Predict *Salary* using *Age* and *MBA* (code as yes $= 1$, no $= 0$)

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

where

Y = salary

X_1 = age

X_2 = MBA indicator (0 or 1)

	A	B	C	D
1	**Employee Salary Data**			
2				
3	**Employee**	**Salary**	**Age**	**MBA**
4	1	$ 28,260	25	No
5	2	$ 43,392	28	Yes
6	3	$ 56,322	37	Yes
7	4	$ 26,086	23	No
8	5	$ 36,807	32	No

A Model with Categorical Variables

$$Salary = 893.59 + 1044.15 \times Age + 14767.23 \times MBA$$

- If $MBA = 0, salary = 893.59 + 1044 \times Age$
- If $MBA = 1, salary = 15{,}660.82 + 1044 \times Age$

	A	B	C	D	E	F	G
1	SUMMARY OUTPUT						
2							
3	*Regression Statistics*						
4	Multiple R	0.976118476					
5	R Square	0.952807278					
6	Adjusted R Square	0.949857733					
7	Standard Error	2941.914352					
8	Observations	35					
9							
10	ANOVA						
11		*df*	*SS*	*MS*	*F*	*Significance F*	
12	Regression	2	5591651177	2795825589	323.0353318	6.05341E-22	
13	Residual	32	276955521.7	8654860.054			
14	Total	34	5868606699				
15							
16		*Coefficients*	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
17	Intercept	893.5875971	1824.575283	0.489751015	0.627650922	-2822.950634	4610.125828
18	Age	1044.146043	42.14128238	24.77727265	1.8878E-22	958.3070599	1129.985026
19	MBA	14767.23159	1351.801764	10.92411031	2.49752E-12	12013.7015	17520.76168

Interactions

- An interaction occurs when the effect of one variable is dependent on another variable.
- We can test for interactions by defining a new variable as the product of the two variables, $X_3 = X_1 \times X_2$, and testing whether this variable is significant, leading to an alternative model.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$$

Incorporating Interaction Terms in a Regression Model

Example: Define an interaction between *Age* and *MBA* and re-run the regression

	A	B	C	D	E
1	**Employee Salary Data**				
2					
3	**Employee**	**Salary**	**Age**	**MBA**	**Interaction**
4	1	$ 28,260	25	0	0
5	2	$ 43,392	28	1	28
6	3	$ 56,322	37	1	37
7	4	$ 26,086	23	0	0

	A	B	C	D	E	F	G
1	SUMMARY OUTPUT						
2							
3	*Regression Statistics*						
4	Multiple R	0.989321416					
5	R Square	0.978756863					
6	Adjusted R Square	0.976701076					
7	Standard Error	2005.37675					
8	Observations	35					
9							
10	ANOVA						
11		*df*	*SS*	*MS*	*F*	*Significance F*	
12	Regression	3	5743939086	1914646362	476.098288	5.31397E-26	
13	Residual	31	124667613.2	4021535.91			
14	Total	34	5868606699				
15							
16		*Coefficients*	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
17	Intercept	3902.509386	1336.39766	2.920170772	0.006467654	1176.908389	6628.110383
18	Age	971.3090382	31.06887722	31.26308786	5.23658E-25	907.9436454	1034.674431
19	MBA	-2971.080074	3026.24236	-0.98177202	0.333812767	-9143.142058	3200.981911
20	Interaction	501.8483604	81.55221742	6.153705887	7.9295E-07	335.5215164	668.1752044

The *MBA* indicator is not significant; drop and re-run

Incorporating Interaction Terms in a Regression Model

Adjusted R^2 increased slightly, and both age and the interaction term are significant. The final model is

$$Salary = 3{,}323.11 + 984.25 \times Age + 425.58 \times MBA \times Age$$

	A	B	C	D	E	F	G
1	SUMMARY OUTPUT						
2							
3	*Regression Statistics*						
4	Multiple R	0.98898754					
5	R Square	0.978096355					
6	Adjusted R Square	0.976727377					
7	Standard Error	2004.24453					
8	Observations	35					
9							
10	ANOVA						
11		*df*	*SS*	*MS*	*F*	*Significance F*	
12	Regression	2	5740062823	2870031411	714.4720368	2.80713E-27	
13	Residual	32	128543876.4	4016996.136			
14	Total	34	5868606699				
15							
16		*Coefficients*	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
17	Intercept	3323.109564	1198.353141	2.773063675	0.009184278	882.1440943	5764.075033
18	Age	984.2455409	28.12039088	35.00113299	4.40388E-27	926.9661791	1041.524903
19	Interaction	425.5845915	24.81794165	17.14826304	1.08793E-17	375.0320986	476.1370843

Categorical Variables with More than Two Levels

- When a categorical variable has $k > 2$ levels, we need to add $k - 1$ additional variables to the model

A Regression Model with Multiple Levels of Categorical Variables

Example: ***Surface Finish*** provides measurement of the surface finish of 35 parts produced on a lathe, along with the revolutions per minute (RPM) of the spindle and one of four types of cutting tools used.

	A	B	C	D
1	Surface Finish Data			
2				
3	Part	Surface Finish	RPM	Cutting Tool
4	1	45.44	225	A
5	2	42.03	200	A
6	3	50.10	250	A
7	4	48.75	245	A
8	5	47.92	235	A
9	6	47.79	237	A
10	7	52.26	265	A
11	8	50.52	259	A
12	9	45.58	221	A
13	10	44.78	218	A
14	11	33.50	224	B
15	12	31.23	212	B
16	13	37.52	248	B
17	14	37.13	260	B
18	15	34.70	243	B

A Regression Model with Multiple Levels of Categorical Variables

Because we have k = 4 levels of tool type, we will define a regression model of the form

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \varepsilon$$

where

Y = *surface finish*

X_1 = *RPM*

X_2 = 1 if tool type is B and 0 if not

X_3 = 1 if tool type is C and 0 if not

X_4 = 1 if tool type is D and 0 if not

	A	B	C	D	E	F
1	**Surface Finish Data**					
2						
3	**Part**	**Surface Finish**	**RPM**	**Type B**	**Type C**	**Type D**
4	1	45.44	225	0	0	0
5	2	42.03	200	0	0	0
6	3	50.10	250	0	0	0
7	4	48.75	245	0	0	0
8	5	47.92	235	0	0	0
9	6	47.79	237	0	0	0
10	7	52.26	265	0	0	0
11	8	50.52	259	0	0	0
12	9	45.58	221	0	0	0
13	10	44.78	218	0	0	0
14	11	33.50	224	1	0	0
15	12	31.23	212	1	0	0
16	13	37.52	248	1	0	0
17	14	37.13	260	1	0	0
18	15	34.70	243	1	0	0
19	16	33.92	238	1	0	0
20	17	32.13	224	1	0	0
21	18	35.47	251	1	0	0
22	19	33.49	232	1	0	0
23	20	32.29	216	1	0	0
24	21	27.44	225	0	1	0
25	22	24.03	200	0	1	0
26	23	27.33	250	0	1	0
27	24	27.20	245	0	1	0
28	25	27.10	235	0	1	0
29	26	27.30	237	0	1	0
30	27	28.30	265	0	1	0
31	28	28.40	259	0	1	0
32	29	26.80	221	0	1	0
33	30	26.40	218	0	1	0
34	31	21.40	224	0	0	1
35	32	20.50	212	0	0	1
36	33	21.90	248	0	0	1
37	34	22.13	260	0	0	1
38	35	22.40	243	0	0	1

A Regression Model with Multiple Levels of Categorical Variables

Regression results:

	A	B	C	D	E	F	G
1	SUMMARY OUTPUT						
2							
3	*Regression Statistics*						
4	Multiple R	0.994447053					
5	R Square	0.988924942					
6	Adjusted R Square	0.987448267					
7	Standard Error	1.089163115					
8	Observations	35					
9							
10	ANOVA						
11		*df*	*SS*	*MS*	*F*	*Significance F*	
12	Regression	4	3177.784271	794.4460678	669.6973322	7.32449E-29	
13	Residual	30	35.58828875	1.186276292			
14	Total	34	3213.37256				
15							
16		*Coefficients*	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
17	Intercept	24.49437244	2.473298088	9.903526211	5.73134E-11	19.44322388	29.54552101
18	RPM	0.097760627	0.010399996	9.400064035	1.89415E-10	0.076521002	0.119000252
19	Type B	-13.31056756	0.487142953	-27.32374035	9.37003E-23	-14.3054462	-12.31568893
20	Type C	-20.487	0.487088553	-42.06011387	3.12134E-28	-21.48176754	-19.49223246
21	Type D	-26.03674519	0.596886375	-43.62094073	1.06415E-28	-27.25574979	-24.81774059

$$Surface\ finish = 24.49 + 0.098\ RPM - 13.31\ type\ B - 20.49\ type\ C - 26.04\ type\ D$$



Regression Models with Nonlinear Terms

Regression Models with Nonlinear Terms

- Curvilinear models may be appropriate when scatter charts or residual plots show nonlinear relationships.
- A second order polynomial might be used

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \varepsilon$$

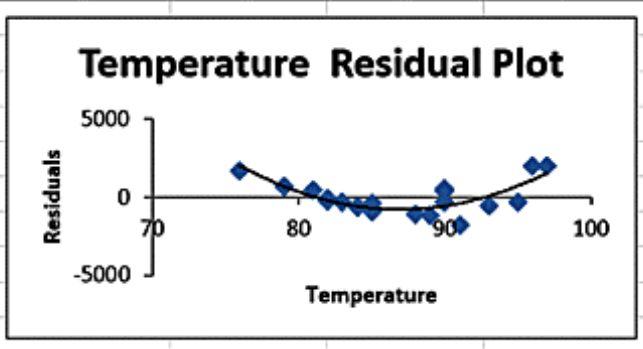
- Here β_1 represents the linear effect of X on Y and β_2 represents the curvilinear effect.
- This model is linear in the β parameters so we can use linear regression methods.

Modelling Beverage Sales using Curvilinear Regression

Example: The U-shape of the residual plot (a second-order polynomial trendline was fit to the residual data) suggests that a linear relationship is not appropriate

	A	B
1	Beverage Sales	
2		
3	Temperature	Sales
4	85	$ 1,810
5	90	$ 4,825
6	79	$ 438
7	82	$ 775
8	84	$ 1,213
9	96	$ 8,692

	A	B	C	D	E	F	G
1	SUMMARY OUTPUT						
2							
3	*Regression Statistics*						
4	Multiple R	0.922351218					
5	R Square	0.850731769					
6	Adjusted R Square	0.842875547					
7	Standard Error	1041.057399					
8	Observations	21					
9							
10	ANOVA						
11		*df*	*SS*	*MS*	*F*	*Significance F*	
12	Regression	1	117362193.6	117362193.6	108.2876347	2.7611E-09	
13	Residual	19	20592209.67	1083800.509			
14	Total	20	137954403.2				
15							
16		*Coefficients*	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
17	Intercept	-32511.24671	3408.723477	-9.53766034	1.12197E-08	-39645.78695	-25376.70648
18	Temperature	408.6026284	39.26555335	10.40613447	2.7611E-09	326.4188807	490.786376

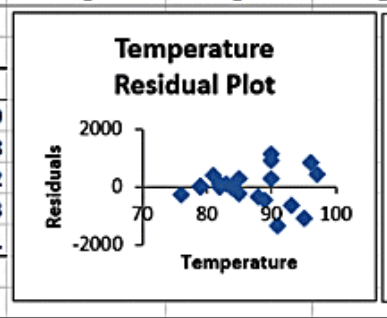


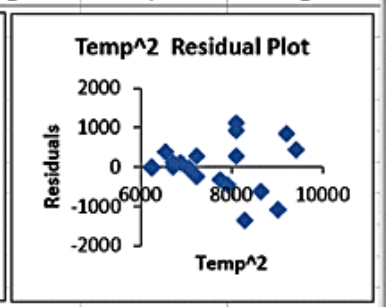
Modelling Beverage Sales using Curvilinear Regression

- Add a variable for temperature squared.
- The model is

$$sales = 142{,}850 - 3{,}643.17 \times temperature + 23.3 \times temperature^2$$

	A	B	C	D	E	F	G
1	SUMMARY OUTPUT						
2							
3	*Regression Statistics*						
4	Multiple R	0.973326989					
5	R Square	0.947365428					
6	Adjusted R Square	0.941517142					
7	Standard Error	635.1365123					
8	Observations	21					
9							
10	ANOVA						
11		*df*	*SS*	*MS*	*F*	*Significance F*	
12	Regression	2	130693232.2	65346616.12	161.9902753	3.10056E-12	
13	Residual	18	7261171.007	403398.3893			
14	Total	20	137954403.2				
15							
16		*Coefficients*	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
17	Intercept	142850.3406	30575.70155	4.672021683	0.000189738	78613.17532	207087.5059
18	Temperature	-3643.171723	705.2304165	-5.165931075	6.492E-05	-5124.805849	-2161.537598
19	Temp^2	23.30035581	4.053196314	5.748637374	1.89343E-05	14.78490634	31.81580528







- Modelling relationships and trends in data
- Simple linear regression
- Residual analysis and regression assumption
- Multiple linear regression
- Building good regression model
- Regression with categorical variables
- Regression models with nonlinear terms